

CSA15 Cloud Computing and Big Data Analytics

CO4 AT1 - Big Data Analytics Class Test Student Template

Course Code	CSA15	Slot	B
Course Title	Cloud Computing and Big Data Analytics	Assessment	CO4 AT1
Assessment Title	Big Data Analytics Class Test	Bloom Level	BL4 - Analyze
Faculty	Dr C. Rajesh Babu	Employee ID	SSETSCS415
Submission Mode	Individual digital answer template through GitHub and Google Classroom	Total Marks	30

Student Details

Student Name	D.S.Tharuniya Priya	Register Number	192511317
Class / Section	CSA1512	Slot	B
Capstone Title	PulsePay : Cloud based Fake Payment Screenshot Detection	Submission Date	19-08-2026
GitHub Repository Link	https://github.com/tharu-ui/Cloud-Computing-and-BDA	Google Classroom Status	Turned in

Assessment Overview

CO4 Focus

Analyze big data characteristics, processing needs, security challenges, business drivers and domain applications. This template includes explanation notes to guide the thinking process. The final answer must be written in the student's own words.

How to use this template:

- Read each question and the explanation notes before answering.
- Use your capstone product as an example wherever relevant.
- Write concise technical answers; avoid copying the explanation notes directly.
- Use diagrams or small tables wherever they improve clarity.
- Upload the completed DOCX/PDF and supporting evidence to GitHub; share the repository link through Google Classroom.

Assessment Rubric

Criteria	Marks	Evaluation Focus	Student Check
Concept Accuracy	9	Correct technical meaning, terminology and examples.	Have I used correct big data terms?

Coverage	6	All required parts of the question are answered.	Have I answered every part?
Examples	4.5	Relevant cloud/capstone/domain examples are included.	Have I used meaningful examples?
Analysis	6	Implications, security, scalability and decision value are explained.	Have I explained why it matters?
Clarity	4.5	Answer is organized, concise and readable.	Is my answer easy to evaluate?

Code of Conduct and Student Assurance

I D.S.Tharuniya Priya(192511317) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: D.S.Tharuniya Priya

Question and Answer Sections

Answer all questions. Each question carries 3 marks. Use the explanation notes for direction, but write the final answer independently.

Question 1 (3 Marks)

Define Big Data and explain why traditional data processing approaches are often insufficient for big data environments.

Explanation Notes

1	Start with a precise definition: Big Data refers to datasets whose size, speed, variety or complexity exceeds the capability of conventional tools.
2	Mention why spreadsheets or single-server databases struggle: storage limits, slow queries, concurrency issues, schema rigidity and limited scalability.
3	Use one cloud/capstone-style example such as app logs, IoT feeds, transactions, images or clickstream events.
4	End with why distributed storage and parallel processing become necessary.

Student Response

Big data refers to the datasets which are too large, fast updating and are too complexly structured to be handled by traditional tools such as Excel. These traditional tools struggle because they are single server databases that have storage limits, slower query processing and limited scalability which can't cater the rapid growing big datasets with such rigid schemas. For example, in PulsePay(Capstone title) thousands of UPI payment screenshots, transaction logs are to be handled everyday which can't be done by using excel and other traditional tools. This is where distributed storage and parallel processing become necessary so they can split the data across multiple machines and process it simultaneously instead of relying on a single machine to do everything.

Question 2 (3 Marks)

Explain the 5Vs of Big Data with one practical example for any two Vs.

Explanation Notes

1	List the 5Vs clearly: Volume, Velocity, Variety, Veracity and Value.
2	For at least two Vs, connect the definition to a concrete scenario instead of writing only textbook meaning.
3	Example: In a smart traffic system, velocity refers to continuous vehicle movement updates; value refers to congestion prediction.
4	Show that each V affects system design, storage, processing or analytics decisions.

Student Response

The five V's of Big data are volume, velocity, variety, veracity and value.

1. Volume refers to the amount of data generated. In PulsePay thousands of payment screenshots get uploaded daily which adds up to the data volume.
2. Velocity refers to the speed at which data is generated and need to be processed. In PulsePay a screenshot must be

verified by cloud vision API almost instantly so a merchant doesn't have to keep waiting to check if the payment was genuinely made.

3.Variety refers to the different formats of data.

4.Veracity refers to the accuracy of the data.

5.Value is the actual usefulness of the data extracted.

Question 3 (3 Marks)

Differentiate structured, semi-structured, and unstructured data with examples.

Explanation Notes

1	Structured data has a fixed schema, such as customer table, transaction table or attendance record.
2	Semi-structured data has tags/keys but flexible structure, such as JSON, XML, API responses or log files.
3	Unstructured data has no fixed tabular form, such as images, videos, audio, PDFs, emails or social media text.
4	Use examples from one cloud-backed application to make the comparison coherent.

Student Response

- 1.Structured data has a fixed format, like a transaction record in PulsePay's Firestore database with fields such as amount, merchant ID, timestamp and status.
- 2.Semi-structured data has tags or keys but no strict tabular form, like the JSON response by the Cloud Vision API after analyzing a screenshot or logs generated by Cloud Functions.
- 3.Unstructured data has no fixed structure at all, like the actual UPI payment screenshot images uploaded by merchants, which are just raw image files with no predefined fields.

Question 4 (3 Marks)

Explain the role of distributed file systems in big data processing.

Explanation Notes

1	State the main need: very large datasets are split and stored across multiple machines.
2	Mention core benefits: scalability, fault tolerance, parallel access and support for batch processing.
3	Use HDFS or cloud object storage as an example, but do not confuse file storage with relational database storage.
4	Connect storage to processing: data blocks can be processed in parallel by frameworks such as MapReduce or Spark.

Student Response

Distributed file system exists in very large databases because these are too enormous to be processed in a single machine and are hence slit into smaller blocks and processed in multiple machines. This increases the scalability ,fault tolerance , parallel access and supports batch processing. For example , HDFS in Hadoop system and PulsePay's cloud storage uses GCP in a similar way in which the screenshots and files are distributed across storage infrastructure rather than a sitting on one server. This storage layer connects to the processing where frameworks like MapReduce or Spark can parallel process the data.

Question 5 (3 Marks)

List any three major security concerns in big data platforms and briefly explain each.

Explanation Notes

1	Choose three concerns such as unauthorized access, data leakage, privacy violation, weak encryption, poor audit logs or insecure APIs.
2	For each concern, explain the risk and not only the keyword.
3	Add one control where possible: role-based access, encryption, anonymization, masking, logging or monitoring.
4	Show awareness that big data combines many data sources, so security and governance become more complex.

Student Response

1. Unauthorized access : since the data is taken from various sources there is a high chance that someone might access the sensitive information by breaching the system. In PulsePay this could be accessing the merchant transaction data. A control here can be role based access so only the authorized personnel can access the data.

2. Data leakage : large volumes of data moving between services increase the chance of accidental exposure , especially with financial data like upi screenshots . Encryption during transit can increase the security.

3. Privacy violation : combining data from various sources can accidentally expose personal information such as upi id etc. Data masking or anonymization can help protect user privacy while still allowing analysis.

Question 6 (3 Marks)

What are data appliances and integration tools? Explain their role in big data workflows.

Explanation Notes

1	A data appliance is a specialized hardware/software platform optimized for storage, processing or analytics workloads.
2	Integration tools move, transform or synchronize data between sources, storage systems and analytics layers.
3	Mention ETL/ELT, API connectors, message queues, ingestion tools or workflow schedulers as examples.
4	Explain the workflow: source data -> ingestion -> cleaning/transformation -> storage -> processing -> dashboard/decision.

Student Response

A data appliance is a specialized hardware or software platform that is pre-optimized for storage, processing or analytics workloads. Integration tools are the tools that move, transform or synchronize data between different sources, storage layers and analytics systems such as ELT/ETL pipelines ,API Connectors, message queues and workflow schedulers. In Plusepay this workflow is like : merchant uploads a screenshot → Cloud Function is triggered (ingestion) → Cloud Vision API extracts and analyzes the image (transformation) → result is stored in Firestore (storage) → the app processes the verification logic → the merchant gets a decision on their dashboard.

Question 7 (3 Marks)

Discuss two major challenges organizations face while implementing big data analytics.

Explanation Notes

1	Pick two strong challenges such as data quality, skill gap, infrastructure cost, privacy compliance, integration difficulty or real-time scalability.
2	Explain the cause and impact of each challenge.
3	Suggest a practical mitigation for each challenge, such as governance, cloud scaling, training, access control or phased implementation.
4	Try to connect the challenge to business impact, not only technical difficulty.

Student Response

Two major challenges are :

Data quality – incoming data can be incomplete, duplicated or inconsistent, which reduces the accuracy of any analysis built on it. In PulsePay , a blurry or partially cropped screenshot could cause the Vision API to misread transaction details, causing wrong fraud predictions. A practical mitigation is setting up validation rules and cleaning steps before data enters the analytics pipeline.

Infrastructure cost and cloud billing – running big data workloads (storage, compute, APIs) at scale can get expensive quickly, especially with pay-as-you-go cloud pricing. This was a real issue faced in PulsePay's own development, where a GCP/Firebase billing wall blocked further testing. A mitigation here is phased implementation and setting budget alerts/quotas so cost is controlled from the start.

Both challenges connect directly to business impact where poor data quality leads to wrong decisions, and uncontrolled costs eat into the ROI that big data adoption is supposed to deliver.

Question 8 (3 Marks)

Write short notes on any one domain application of big data analytics.

Explanation Notes

1	Select one domain: healthcare, retail, agriculture, finance, education, smart city, transport or cybersecurity.
2	Identify data sources in that domain.
3	Explain how analytics converts raw data into decisions: prediction, recommendation, alert, optimization or risk detection.
4	Mention one expected outcome such as reduced waiting time, fraud detection, crop advisory or improved user experience.

Student Response

Domain: Finance / Digital Payments (Fraud Detection).

Data sources in this domain include transaction records, UPI payment screenshots, merchant activity logs and device/location metadata. Analytics converts this raw data into decisions by running automated checks for example, PulsePay uses the Cloud Vision API to analyze uploaded UPI screenshots and detect signs of tampering or fake payment confirmations, then flags suspicious transactions for the merchant in near real time. This is essentially fraud detection applied to the payments domain. The expected outcome is a reduction in financial loss for small merchants who might

otherwise accept a fake payment screenshot as proof of payment, along with faster verification compared to manual checking.

Question 9 (3 Marks)

How do business drivers influence the adoption of big data solutions?

Explanation Notes

1	Business drivers are reasons that push an organization to adopt big data, such as cost reduction, personalization, speed, risk control or better decisions.
2	Explain how each selected driver creates a technical requirement.
3	Example: personalization requires behavior tracking and recommendation analytics; risk control requires fraud detection and monitoring.
4	Conclude that big data adoption must be justified by measurable business value.

Student Response

Business drivers are the practical reasons that push an organization to adopt big data solutions like cost reduction, personalization, speed of operations, risk control or better decision-making. Each driver creates a specific technical requirement. For example, risk control (fraud prevention) is the main driver behind PulsePay, it requires image analysis (Cloud Vision API), real-time processing (Cloud Functions) and secure data storage (Firestore) to actually deliver that goal. Similarly, a driver like personalization in an e-commerce app would require behavior tracking and recommendation analytics. big data adoption is not just a technical upgrade it has to be justified by measurable business value, like fewer fraud cases or faster merchant payouts.

Question 10 (3 Marks)

Explain the importance of big data analytics in decision making.

Explanation Notes

1	Explain that analytics helps convert raw, high-volume data into useful patterns, predictions, alerts or recommendations.
2	Differentiate descriptive, diagnostic, predictive and prescriptive decision support if space permits.
3	Use one example: dashboard for hospital beds, forecast for demand, fraud alert for bank, traffic prediction for smart city.
4	End with the idea that decisions become faster, evidence-based and scalable.

Student Response

Big data analytics is important in decision making because it converts raw, high-volume data into useful patterns, predictions, alerts and recommendations that humans can act on quickly. Decision support can be broken into four types: descriptive (what happened), diagnostic (why it happened), predictive (what might happen) and prescriptive (what should be done about it). For example, in PulsePay, when a merchant uploads a screenshot, the system doesn't just store the image. it analyzes it and gives an immediate decision (genuine or fake), which is essentially predictive analytics

happening in real time. This is similar to a fraud alert system in a bank or a bed-availability dashboard in a hospital. Overall, big data analytics makes decisions faster, more evidence-based, and easier to scale across large numbers of users or transactions instead of relying on manual review.

Submission Checklist

Checklist Item	Status	Remarks
Student details completed.	Done / Pending	Done
All 10 questions answered.	Done / Pending	Done
Examples are connected to cloud/capstone/domain context.	Done / Pending	Done
Security/privacy points are included wherever relevant.	Done / Pending	Done
Document exported as PDF if requested.	Done / Pending	Done
GitHub link shared through Google Classroom.	Done / Pending	Done